Photon-Ring Retarded-Time Tomography I: Class-Dependent Identifiability and the Separation of Historical Reach from Algebraic Rank in Near-Critical Null Geodesics

Preprint draft — theory and controlled synthetic computation only. No telescope detection, no laboratory result, and no recovery from a resolved real photon ring is claimed anywhere in this manuscript.

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Abstract

Light that passes near the photon shell of a Kerr black hole reaches a distant observer along a family of increasingly delayed null geodesics, so a single image contains a superposition of source epochs. We ask what of that history is actually recoverable, and we separate three questions that are routinely conflated: whether a declared source model is identifiable, how far back in retarded time an observation can see, and how strongly higher-order images are suppressed.

We build a matrix-free whitened forward operator directly from per-ray Kerr transfer maps on 12 registered spin–inclination geometries, retaining orders n = 0, 1, 2. The measurement model is pixel-integrated with white noise of fixed density per unit solid angle, so the whitened row carries `sqrt(dOmega) g^3`; an earlier convention using `g^3` with a flat per-row noise made Fisher information scale with pixel count and is retired, with the correction and everything it moved recorded rather than absorbed.

Three results. First, **historical extension is real but modest and set by inclination, not spin**: the resolved stack sees deeper than the direct image at 12 of 12 geometries, with median recoverable depth 60, 84 and 144 M at i = 20, 50 and 75 degrees, and the four spins give a single value at every inclination. Second, **retarded-time diversity supplies more of that reach than spatial remapping, but not all of it**: measured against the direct image as the zero point, flattening the delays costs a median 0.98 of the higher orders' entire contribution while transplanting the spatial map costs 0.57. Third, and most consequential for how such claims should be stated, **identifiability and historical reach are different quantities that move independently**. Enriching the declared source class from 224 to 1056 dimensions drives the resolved operational-rank fraction down by 16.8 percentage points and the smallest determined singular value down by five orders of magnitude, while the deepest recoverable epoch moves by at most 1 grid step of 4 M. Full column rank on the registered class is therefore a statement about that class and not about the continuum: temporal enrichment alone exposes a direct-channel null space of up to 100 dimensions.

We report a negative control throughout. Permuting each order's delays, positions and weights independently — preserving all three marginals and destroying only their pairing — yields a better-conditioned operator than the physical one at 12 of 12 geometries across 16 frozen seeds. Conditioning is not evidence of physical content, and no conclusion here rests on it.

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1. Introduction

A photon emitted from the equatorial plane near a Kerr black hole can reach a distant observer directly, or after one or more near-critical windings around the photon shell. Each additional half-orbit adds delay, so the n-th image of a time-varying source shows that source as it was further in the past. This is the observation behind every proposal to read accretion history out of a resolved photon ring.

Whether that history can be *recovered* is a different question from whether it is *present*, and the difference is where most of the difficulty lives. Three quantities are easy to conflate:

1. **Algebraic identifiability** — does the forward operator have a trivial null space on the declared source model? This is a property of the pair (operator, model class), and a small enough class is identifiable under almost any operator.
2. **Historical reach** — how far into the past can a localized change in the source be detected at a given signal-to-noise ratio? This is a property of where the operator's *sensitivity* lives in retarded time.
3. **Throughput suppression** — how much fainter is each successive image? This is a property of the transfer coefficients alone and says nothing directly about either of the first two.

This paper measures all three on validated Kerr and Schwarzschild ray maps and shows that they separate. It also documents a measurement-convention defect we found and corrected mid-campaign, because several of the conclusions we would otherwise have reported were artefacts of it.

1.1 What is deliberately not claimed

No statement here concerns a real observation. The source model is a declared finite-dimensional class, and no result is extrapolated from full rank on such a class to injectivity on a continuum of source functions — that inference is exactly what section 7 shows to be invalid. No machine-learned reconstruction is performed. No geometry-mismatch or order-leakage study is included. The recoverable depth reported everywhere is a detection statement about a localized unit-norm historical mode, never the largest delay any ray happens to carry.

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2. Forward model and measurement convention

2.1 The operator

Each retained ray p of image order n lands at an equatorial source point `(r, phi)` and carries a delay `Delta t` and a redshift factor `g`. The observation at observer time `t_o` is built row by row rather than from an order-wide delay:

```
z_{n,p}(t_o) = dOmega_{n,p} g_{n,p}^3 j(r_{n,p}, phi_{n,p}, t_o - Delta t_{n,p}) + eta
Var(eta)     = sigma_Omega^2 dOmega_{n,p}
```

The distinction from a delay ladder matters. The pilot geometry's orders span *overlapping* retarded windows, so an image order does not correspond to one source epoch, and `a_n j(t_o - n tau)` is only an asymptotic summary. A distributed delay kernel and a delay ladder have different null spaces.

Whitening under the declared noise gives the row that every audit consumes:

```
Atilde = sqrt(dOmega) / sigma_Omega * g^3 * B(r, phi, t)
```

2.2 The square root is load-bearing

An earlier revision of this work used `c = g^3` with a flat per-row noise. That convention makes the Fisher information scale with the *number of rows*: splitting one pixel into k equal-area children carrying the same transfer value multiplies the Gram matrix by k. Adaptive ray counts differ by an order of magnitude across geometry and order, and the lensing bands differ in solid angle by roughly three orders of magnitude, so the convention silently reweighted the bands against one another — handing the thin, deep, high-order bands a far quieter detector per unit sky than the direct image.

The corrected convention is invariant under the same split-and-merge test to 5.4e-15, against 7 for the retired one, and the check is a registered gate. Every result in this manuscript is computed under the corrected convention; section 9 lists what the correction moved.

Derived arms are linear maps of the same resolved data with their covariance propagated, never separate models with their own noise level:

```
unresolved image   y_U = L y_R,  C_U = L C_R L^T
total flux         y_F = S y_R,  C_F = S C_R S^T
```

with `C_R = sigma_Omega^2 diag(dOmega)`. One noise density is fixed once from the direct arm's clean response to the declared reference source and shared by every arm. No arm may choose its own sigma; an arm that did would be measuring its own row count.

2.3 Source classes and the localized historical probe

The registered primary class **C224** is 4 radial cubic B-splines (uniform knots in log r) x 7 real azimuthal Fourier modes x 8 global temporal DCT modes = 224 dimensions. Every temporal factor spans the whole history, so C224 answers "can a low-dimensional global model be fitted" and cannot be indexed by epoch.

Epoch-resolved questions use the **localized class**: the same radial and azimuthal factors crossed with one compact Gaussian bump in retarded age of half width 3 M. Its m = 0, partition-of-unity contraction is a spatially flat scalar probe, and both are used — the scalar probe for the registered depth and innovation statistics, the full 28-dimensional class where a spatially flat probe would be blind (section 5.2).

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3. Geometries, ray maps, and the common age grid

Twelve registered geometries span spin a* in (0, 0.5, 0.9, 0.98) and inclination in (20, 50, 75) degrees, with orders n = 0, 1, 2 and 1536 rays retained per order, stratified in screen azimuth, source radius and delay with each band's total solid angle preserved. Observations are 8 times over 20 M.

At exactly zero spin the pinned tracer packages are unusable — one package's critical-curve parameterization is singular there and the other's default Keplerian branch sets the orbital angular velocity to zero — so an explicitly validated Schwarzschild backend is used, and the orbital orientation at a* = 0 is the a -> 0+ disk-rotation branch. A Schwarzschild black hole has no preferred spin direction, but the disk still has an orbital orientation, and matching the positive-spin limit keeps the grid continuous.

3.1 Delay summaries that converge

The sampled maximum ray delay is an extreme-value statistic set by whichever single ray sits closest to a band edge. It does not converge under refinement and is not used as a historical depth anywhere in this work. The converging summaries are weighted quantiles under three weightings: solid angle (`dOmega`), throughput (`dOmega g^3`), and Fisher (`dOmega g^6`, the squared whitened row weight). Only the first two are properties of the spacetime and the transfer alone; the third depends on the declared measurement model and is labelled as such.

The common age ceiling is fixed from those source-independent summaries over the *whole* grid before any operator is evaluated:

```
A_max = T_obs + 1.25 max_(g,n) {Q_0.999^Omega, Q_0.999^I} + 2h
      = 249.6 M  ->  252 M after rounding up to the age step
```

with the maximum, 178.9 M, occurring at `a098_i075`. Because the ceiling sits above the deepest ray any geometry carries, 0 of 1152 depth entries in the whole campaign are right-censored: the depths reported below are measurements, not grid ceilings. For contrast, the largest sampled maximum delay anywhere on the grid is 191.5 M — the statistic we do not use.

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4. Correctness

Every mechanical gate is the worst case over all 12 geometries (E3C) or all class–anchor combinations (E3D).

gate	status	measured	threshold
`E3C_G2_physical_dense_matrix_free`	**PASS**	0	1e-10
`E3C_G3_physical_adjoint`	**PASS**	2.67e-13	1e-08
`E3C_G4_physical_resolved_unresolved_mixing`	**PASS**	0	1e-10
`E3C_G4b_linear_collapse_covariance_propagation`	**PASS**	0	1e-12
`E3C_G6_physical_Gram_monotonicity`	**PASS**	1.86e-16	1e-10
`E3C_G6b_resolved_dominates_direct`	**PASS**	1.57e-16	1e-10
`E3C_G9w_weight_semantics`	**PASS**	4.38e-16	1e-10
`E3C_freeze_raymap_hashes`	**PASS**	0	0
`E3C_frozen_grid_invariance`	**PASS**	dims=[224] n_ages=[64]	dims=[224] n_ages=[64]
`E3C_H2_registered_statistic_is_an_identity`	**PASS**	12	12
`E3D_adjoint`	**PASS**	4.89e-13	1e-08
`E3D_dense_smoke_comparison`	**PASS**	0	1e-10
`E3D_Gram_monotonicity`	**PASS**	1.14e-16	1e-10
`E3D_class_nesting`	**PASS**	2.73e-14	1e-10
`E3D_enrichment_does_not_lose_rank`	**PASS**	0	0
`G10q_continuum_noise_quadrature_invariance`	**PASS**	5.4e-15	1e-10

`E3C_freeze_raymap_hashes` re-checks every ray map against the digest pinned before evaluation; `E3C_frozen_grid_invariance` checks that the class dimension and age grid were identical at every geometry. `E3D_class_nesting` is discussed in section 7.

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5. Results

5.1 Historical extension is real, modest, and set by inclination

At the reference SNR the resolved stack sees deeper than the direct image at 12 of 12 geometries, and carries strictly positive historical innovation beyond the direct channel's own 99.9% throughput-weighted age boundary at 12 of 12.

Recoverable depth, resolved stack (M):

a* \ i	20°	50°	75°	trend in inclination
0.00	60	84	144	nondecreasing
0.50	60	84	144	nondecreasing
0.90	60	84	144	nondecreasing
0.98	60	84	144	nondecreasing
trend in spin	constant	constant	constant	

Recoverable depth, direct image alone (M):

a* \ i	20°	50°	75°	trend in inclination
0.00	20	60	140	nondecreasing
0.50	20	60	140	nondecreasing
0.90	20	60	140	nondecreasing
0.98	20	60	140	nondecreasing
trend in spin	constant	constant	constant	

The surface is flat in spin and steep in inclination. At each of the three inclinations the four spins give a single value of the resolved depth (1, 1 and 1 distinct values respectively); the medians are 60, 84 and 144 M for the resolved stack against 20, 60 and 140 M for the direct image. Whatever sets historical reach in this operator, it is the geometry of the retarded windows as seen from a given viewing angle, not the spin parameter.

Depth is a threshold statement, so we report alongside it the threshold-independent **historical innovation**

```
J_old = integral over a > A_0,0.999 of log(1 + I(a)) da
```

which integrates information about the localized epoch over exactly the region where the direct channel is essentially absent.

Historical innovation, resolved stack:

a* \ i	20°	50°	75°	trend in inclination
0.00	67.86	40.93	12.36	nonincreasing
0.50	69.94	41.21	15.88	nonincreasing
0.90	70.81	40.44	16.07	nonincreasing
0.98	71.53	40.51	16.22	nonincreasing
trend in spin	nondecreasing	nonmonotone	nondecreasing	

Historical innovation, direct image alone:

a* \ i	20°	50°	75°	trend in inclination
0.00	15.02	8.34	5.07	nonincreasing
0.50	15.28	8.78	4.92	nonincreasing
0.90	15.37	8.24	5.09	nonincreasing
0.98	15.40	7.99	5.24	nonincreasing
trend in spin	nondecreasing	nonmonotone	nonmonotone	

`J_old` falls with inclination while depth rises with it. The two are not in tension: at high inclination the direct channel already reaches far back, so the *additional* history the higher orders supply is a smaller increment even though the absolute reach is larger.

The incremental Gram operator `delta_G = G_resolved - G_direct` separates information reweighted inside the direct channel's support from genuinely new historical information beyond it. Its rank runs from 195 to 224 of 224:

a* \ i	20°	50°	75°	trend in inclination
0.00	224	224	195	nonincreasing
0.50	224	224	196	nonincreasing
0.90	224	224	196	nonincreasing
0.98	224	224	197	nonincreasing
trend in spin	constant	constant	nondecreasing	

5.2 A registered statistic that could not answer its own question

The registered mechanism test compares the full and substituted age-information curves through a relative discrepancy `D`. Applied to the delay-only substitution it is **identically zero at every geometry**, with a maximum over the whole grid of 0.0e+00.

This is an algebraic identity, not a finding. The registered probe is spatially flat, and the delay-only substitution replaces only `source_r` and `source_phi`; the scalar curve therefore *cannot* depend on the substitution. We verified the curves are bitwise identical at all 12 geometries and locked that fact behind a gate, so the zero can never be re-read as support for the delay mechanism. Reporting it as such would have been circular.

The literal values are preserved on the record. Amendment 002 adds the comparison that can discriminate: the age-resolved information matrix on the registered 28-dimensional localized class, each probe normalised to unit L2 norm over the emission region, compared through the log information volume `sum_k log(1 + SNR^2 lambda_k(a))`.

5.3 Delay diversity supplies more of the reach than spatial remapping

Two substitution arms isolate the candidate mechanisms while holding the measurement model, the noise, the source class and the ray weights fixed. `DELAY_ONLY` keeps each order's physical delays and takes the direct order's spatial map; `SPATIAL_ONLY` keeps each order's spatial map and flattens every delay onto the direct order's.

geometry	D_delay (registered)	D_delay	D_spatial	D_direct	D_delay / D_direct	D_spatial / D_direct
`a000_i020`	0.0e+00	0.1691	0.3108	0.3147	0.537	0.988
`a000_i050`	0.0e+00	0.1512	0.2060	0.2307	0.655	0.893
`a000_i075`	0.0e+00	0.1651	0.3036	0.2294	0.720	1.324
`a050_i020`	0.0e+00	0.1469	0.3229	0.3276	0.448	0.986
`a050_i050`	0.0e+00	0.1685	0.2041	0.2475	0.681	0.825
`a050_i075`	0.0e+00	0.2747	0.3418	0.2663	1.032	1.283
`a090_i020`	0.0e+00	0.1749	0.3344	0.3400	0.514	0.983
`a090_i050`	0.0e+00	0.1613	0.2183	0.2688	0.600	0.812
`a090_i075`	0.0e+00	0.2147	0.3099	0.3074	0.699	1.008
`a098_i020`	0.0e+00	0.1653	0.3386	0.3457	0.478	0.979
`a098_i050`	0.0e+00	0.1187	0.2416	0.2984	0.398	0.810
`a098_i075`	0.0e+00	0.1288	0.3734	0.3379	0.381	1.105

The direct arm's discrepancy is the natural zero point: it is what remains when the higher orders are removed altogether. Normalised by it, flattening the delays costs a median 0.98 of everything the higher orders contribute — essentially all of it — while transplanting the spatial map costs 0.57. Delay-only is the closer of the two substitutions at 12 of 12 geometries.

This is weaker than a single-geometry canary suggested. Under the scalar statistic the delay-only arm looked like an exact reproduction of the full operator. On the localized class it is a median 0.165 relative departure against the direct arm's 0.303. Retarded-time diversity carries the larger share of the historical reach; it does not carry all of it, and spatial remapping is not negligible.

Conditioning is reported beside the discrepancy deliberately: spatial remapping can improve `kappa+` without moving the oldest detectable epoch, so reading the mechanism off a depth endpoint alone would credit delay with work that spatial diversity does.

5.4 What the order labels are worth

`UNRESOLVED_IMAGE` sums the orders into a single image plane and pays the summed noise through `C_U = L C_R L^T`.

geometry	T(unresolved) / T(resolved)	J_old(unresolved) / J_old(resolved)
`a000_i020`	0.333	0.243
`a000_i050`	0.714	0.235
`a000_i075`	0.972	0.439
`a050_i020`	0.333	0.241
`a050_i050`	0.714	0.245
`a050_i075`	0.972	0.381
`a090_i020`	0.333	0.244
`a090_i050`	0.714	0.240
`a090_i075`	0.972	0.382
`a098_i020`	0.333	0.243
`a098_i050`	0.714	0.233
`a098_i075`	0.972	0.393

Depth degrades gracefully — the ratio runs from 0.33 to 0.97, reaching 0.97 at high inclination where the direct channel already dominates the reach. Historical innovation does not: the ratio runs 0.23 to 0.44. Explicit order labels are close to dispensable for *how far back* one can see and load-bearing for *how much* is learned about that history.

5.5 Throughput suppression and sensitivity suppression are different exponents

Comparing orders requires matched support. Each order is sampled at the same fractional position within its own retarded window, giving

```
Gamma_sensitivity_matched = -0.5 log( I_{n+1}^matched / I_n^matched )
```

against the throughput exponent `Gamma_amp = -log(A_g ratio)`. All 19 window fractions are retained at every geometry.

geometry	pair	defined / 19	median	IQR	Gamma_amp	difference	window overlap	disposition
`a000_i020`	0->1	19	2.614	0.397	4.714	2.100	0.000	**undefined**
`a000_i020`	1->2	19	1.831	0.124	3.602	1.771	0.534	supported
`a000_i050`	0->1	19	2.441	0.583	4.274	1.833	0.115	supported
`a000_i050`	1->2	19	2.187	0.263	3.887	1.700	0.493	supported
`a000_i075`	0->1	19	1.979	0.909	3.248	1.269	0.308	supported
`a000_i075`	1->2	19	2.059	0.131	4.057	1.998	0.578	supported
`a050_i020`	0->1	19	2.723	0.126	4.681	1.958	0.000	**undefined**
`a050_i020`	1->2	19	1.794	0.018	3.501	1.706	0.545	supported
`a050_i050`	0->1	19	2.486	0.561	4.272	1.786	0.114	supported
`a050_i050`	1->2	19	2.120	0.157	3.784	1.664	0.521	supported
`a050_i075`	0->1	19	1.530	0.815	2.870	1.340	0.309	supported
`a050_i075`	1->2	19	2.569	0.343	4.388	1.819	0.519	supported
`a090_i020`	0->1	19	2.609	0.259	4.620	2.010	0.000	**undefined**
`a090_i020`	1->2	19	1.704	0.112	3.203	1.499	0.545	supported
`a090_i050`	0->1	19	2.492	0.568	4.249	1.758	0.117	supported
`a090_i050`	1->2	19	1.950	0.211	3.548	1.599	0.557	supported
`a090_i075`	0->1	19	1.560	0.884	2.875	1.315	0.309	supported
`a090_i075`	1->2	19	2.310	0.403	4.251	1.941	0.571	supported
`a098_i020`	0->1	10	1.909	3.409	4.592	2.683	0.435	**undefined**
`a098_i020`	1->2	19	1.682	0.042	3.075	1.394	0.536	supported
`a098_i050`	0->1	19	2.679	1.197	4.230	1.551	0.139	supported
`a098_i050`	1->2	19	1.684	0.627	3.356	1.672	0.724	**undefined**
`a098_i075`	0->1	19	1.611	1.013	2.872	1.262	0.304	supported
`a098_i075`	1->2	19	2.230	0.261	4.120	1.889	0.559	supported

5 of 24 cells are marked `UNDEFINED_NO_COMMON_MATCHED_SUPPORT` and are named rather than dropped:

geometry	pair	reasons
`a000_i020`	0->1	windows_disjoint
`a050_i020`	0->1	windows_disjoint
`a090_i020`	0->1	windows_disjoint
`a098_i020`	0->1	fractions_without_information, negative_exponent_fractions
`a098_i050`	1->2	negative_exponent_fractions

The zero-overlap cells are the substantive caveat. At i = 20 degrees the n = 0 and n = 1 retarded windows are disjoint, so sampling both at the same fractional position compares two epochs that share no support at all; the statistic is still computed and reported, but "matched support" is not what it means there. These cells remain in every denominator: the interpretable set is 19 of 24, not 19 of 19.

On the 19 interpretable cells the sensitivity exponent is below the throughput exponent in all 19, with median difference 1.700 and range 1.262 to 1.998. A single scalar attenuation exponent describes the flux and misdescribes the information.

At the canary geometry, the corrected medians are 2.486 (0 to 1) and 2.120 (1 to 2) against throughput exponents 4.27 and 3.78. The same quantities computed independently in the single-geometry canary tables give 2.486 and 2.120. The corrected gap is close to a convention-level identity: whitened rows carry `sqrt(dOmega)` where flux carries `dOmega`, so information scales linearly in solid angle where throughput scales quadratically, and the exponent difference is approximately half the log solid-angle demagnification. **No asymptotic law is fitted**: n = 0, 1, 2 does not determine one.

5.6 The geometry surface

These are twelve deterministic registered geometries, not a sample from a population. No p-value, confidence interval or significance claim appears anywhere in this paper; the surface is reported cell by cell.

Operational rank, resolved stack, of 224:

a* \ i	20°	50°	75°	trend in inclination
0.00	203	201	185	nonincreasing
0.50	203	201	184	nonincreasing
0.90	203	203	189	nonincreasing
0.98	203	205	193	nonmonotone
trend in spin	constant	nondecreasing	nonmonotone	

Operational rank, direct image, of 224:

a* \ i	20°	50°	75°	trend in inclination
0.00	133	153	159	nondecreasing
0.50	134	153	157	nondecreasing
0.90	135	154	157	nondecreasing
0.98	133	153	156	nondecreasing
trend in spin	nonmonotone	nonmonotone	nonincreasing	

Conditioning `kappa+`, resolved stack:

a* \ i	20°	50°	75°	trend in inclination
0.00	7.04e+05	7.03e+05	2.88e+07	nonmonotone
0.50	7.12e+05	7.04e+05	1.73e+07	nonmonotone
0.90	8.56e+05	6.98e+05	1.36e+07	nonmonotone
0.98	1.19e+06	2.00e+05	7.53e+06	nonmonotone
trend in spin	nondecreasing	nonmonotone	nonincreasing	

Read off the cells rather than summarised: the resolved operational rank is constant at i = 20, nondecreasing at i = 50, nonmonotone at i = 75; nonincreasing at a = *0.00, nonincreasing at a* = 0.50, nonincreasing at a = *0.90, nonmonotone at a* = 0.98. There is no clean monotone spin trend, and section 6.2 shows that what trend there is in rank is partly confounded with the source domain and must not be read as a pure spacetime effect. The inclination dependence is the robust one, and it runs opposite to depth: higher inclination buys reach and costs identifiable directions.

5.7 Arms at a glance

arm	operational rank (median)	range	kappa+ (median)	J_old (median)	T_rec (median)
`DIRECT_PHYSICAL`	153	133–159	9.00e+08	8.29	60
`RESOLVED_PHYSICAL`	202	184–205	7.84e+05	40.72	84
`UNRESOLVED_IMAGE`	178	173–183	9.52e+06	9.66	60
`TOTAL_FLUX`	13	12–14	9.86e+08	6.90	56
`DELAY_ONLY`	212	193–216	9.50e+04	40.72	84
`SPATIAL_ONLY`	178	140–209	9.76e+06	8.33	60
`EQUALIZED_ORDER_SENSITIVITY`	216	196–222	5.04e+04	306.70	120
`PAIRING_DESTROYED`	221	217–224	1.80e+04	28.72	72

Two rows in that table are not measurement architectures and must not be read as ones. `EQUALIZED_ORDER_SENSITIVITY` removes the physical attenuation between orders by fiat; its large `J_old` and depth are an oracle bound on what attenuation costs, not an achievable configuration. `PAIRING_DESTROYED` is the nonphysical negative control of section 6.1. `TOTAL_FLUX` is a genuine but deliberately impoverished readout: it collapses all spatial information and reaches a median operational rank of 13 against the resolved stack's 202, and is included as a floor.

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6. Controls

6.1 A physically meaningless operator is better conditioned than the real one

`PAIRING_DESTROYED` permutes delay, spatial position and quadrature weight independently within each order. All three marginals survive; only their pairing is destroyed. Over 16 frozen seeds per geometry it reaches a median operational rank of 220 and a worst-case conditioning of 5.81e+04, against the physical resolved operator's median 7.84e+05. At 12 of 12 geometries the *worst* seed is still better conditioned than the physical operator.

This is a nonphysical control and is never ranked as an alternative measurement architecture. Its purpose is to refute a specific inference: any argument that reads good conditioning, high operational rank or a large stable rank as evidence that an operator is capturing physical structure is refuted by these rows.

6.2 The source domain moves with spin, and we measured how much

The registered radial support is geometry-dependent — the knots follow the rays, and a higher-spin geometry's rays reach closer to the horizon. A spin trend in any rank statistic is therefore partly a statement about the source domain. Rather than assume the confound away, we repeated three anchor geometries on one fixed radial interval in r/M with identical knot locations.

Operational rank moves under the common support in 12 of 24 anchor–arm combinations, by at most 5 of 224. Recoverable depth is unchanged in 23 of 24. Read strictly: the spin trends in operational rank are partly confounded with the source domain; the depth and innovation results are not.

Separately, near-horizon *ray* coverage does not imply that the emissivity model physically emits to the horizon. Ray-map support and assumed source emission are different objects and are kept apart throughout.

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7. Full rank on a declared class is a statement about that class

The registered class C224 reaches full column rank under 21 of 21 arm–anchor combinations. That fact is often where such analyses stop. It should not be, because it is a joint property of the operator and a 224-dimensional model, and the way to find out how much of it is the model's poverty is to enrich the model.

We do so along a nested ladder on three anchor geometries. The azimuthal and temporal factors are literal prefixes, so their columns are preserved; the radial cubic B-spline factor is refined, so its columns move while its span is contained in the enriched span, to 2.7e-14. Rank statements depend on the span, and that is what the nesting gate checks. Exact adjoint, Gram monotonicity and a dense smoke comparison were run at every class including the 1056-dimensional one; the worst dense-versus-matrix-free discrepancy is 0.0e+00.

class	radial	azimuthal	temporal	dimension	resolved operational rank	as a fraction	sigma_min+
`C224`	4	7	8	224	201	0.897	1.63e-02
`C448_T`	4	7	16	448	365	0.815	7.16e-05
`C528_S`	6	11	8	528	441	0.835	2.87e-04
`C1056_ST`	6	11	16	1056	770	0.729	1.33e-07

Rank deficiency by arm and class. Each cell gives how many of the three anchor geometries are rank-deficient and the range of nullities among those that are; `full` means every anchor reaches full column rank. Aggregating to a single worst nullity would report a deficiency for an arm that is clean at two anchors out of three, which is a different finding:

arm	`C224` (224)	`C448_T` (448)	`C528_S` (528)	`C1056_ST` (1056)
`DIRECT_PHYSICAL`	full	**3/3, −4 to −100**	**1/3, −6**	**3/3, −109 to −265**
`RESOLVED_PHYSICAL`	full	**1/3, −1**	full	**3/3, −11 to −35**
`UNRESOLVED_IMAGE`	full	**1/3, −1**	full	**3/3, −14 to −35**
`DELAY_ONLY`	full	**1/3, −1**	full	**1/3, −36**
`SPATIAL_ONLY`	full	**2/3, −23 to −91**	full	**2/3, −60 to −228**
`EQUALIZED_ORDER_SENSITIVITY`	full	**1/3, −1**	full	**3/3, −4 to −25**
`PAIRING_DESTROYED`	full	full	full	full

Four findings.

Temporal enrichment alone breaks it. Doubling the temporal resolution while holding the spatial factors fixed leaves 9 of the 21 arm–anchor combinations rank-deficient, with the direct channel deficient at every anchor and losing up to 100 dimensions of 448. Nothing about the geometry changed; only the question being asked of it.

Spatial enrichment is far gentler. At C528_S only 1 of the 21 combinations are deficient — the direct channel at a single anchor, by 6 dimensions of 528. The operator resolves the declared spatial class much better than the declared temporal one — the same asymmetry the mechanism decomposition found from the other side.

At the richest class the full physical operator has a null space. 15 of the 21 combinations are deficient at C1056_ST, including the resolved stack itself (up to 35 dimensions of 1056) and the direct channel (up to 265).

The negative control never loses rank. `PAIRING_DESTROYED` reaches full column rank at every class and every anchor, including the 1056-dimensional one, where the physical resolved operator does not. An operator with no physical content is the only one on the table that stays identifiable under enrichment. This is the same lesson as section 6.1 arriving from the rank side: identifiability is not a measure of physical fidelity.

The delay-only equivalence does not survive enrichment. Indistinguishable from the resolved operator by rank on C224, the two separate once the model is rich enough — and not always in the direction one would guess, because substituting the direct order's well-sampled spatial map onto every order is not a strict impoverishment. `DELAY_ONLY` is a mechanism probe, not a measurement architecture.

7.1 Identifiability and reach move independently

Over the same ladder the resolved *operational*-rank fraction falls by 16.8 percentage points and the smallest determined singular value falls from 1.63e-02 to 1.33e-07 — five orders of magnitude. Recoverable depth moves in 7 of 18 physical anchor–arm rows and never by more than 1 grid step of 4 M.

This is the cleanest statement the campaign supports. **A rank statement is not a depth statement, in either direction, and neither is evidence for the other.** Enriching a source model exposes directions the operator cannot determine; it barely changes how far back in retarded time the operator can see. Papers that report identifiability and papers that report historical reach are reporting different physics, and a result in one does not transfer to the other.

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8. Discussion

The picture that survives all of the above is narrower than the one a single-geometry study would support, and we think it is the useful one.

Near-critical null geodesics do create a long, distributed, overlapping retarded-time transfer structure. That structure is not a delay ladder: orders share retarded windows, each ray carries its own delay, and treating the n-th image as a snapshot of one past epoch is an asymptotic convenience that the operator's null structure does not respect.

Higher-order images are drastically suppressed in integrated brightness and retain disproportionate — though not, as we first reported, dramatically disproportionate — Fisher sensitivity to historical source modes. The gap between the throughput and sensitivity exponents is real, roughly a factor of two in the exponent on the interpretable cells, and is largely accounted for by the solid-angle scaling that whitening imposes rather than by anything distinctive about near-critical geodesics.

Recoverable history is controlled by an information-weighted temporal kernel. It is not controlled by the oldest ray, which is an extreme-value statistic that does not converge; not by total photon-ring flux, which the equalized-sensitivity arm shows to be a poor proxy; and not by algebraic rank, which section 7 shows can move by hundreds of dimensions while depth moves by one grid step.

The most transferable methodological point is negative. Three of the quantities one would naturally reach for — total information, algebraic rank, conditioning — each fail as proxies for recoverable history in a specific, demonstrable way: total information is dominated by the bright direct image and barely moves when the informative deep orders are added; rank is a property of the declared class; and conditioning is optimised by a permutation with no physical content at all.

9. What the measurement-model correction changed

The defect and its consequences are recorded in the invalidation ledger as `D-H_flat_sigma_measurement_convention`, and 33 artifacts produced under the retired convention are marked `SUPERSEDED_MEASUREMENT_MODEL_DEFECT`. No table in this manuscript is built from any of them; the builder reads its inputs through the canonical freeze and raises on a superseded path.

What moved, at the canary geometry. The retired-convention values in the middle column are quoted from the defect record and from the pre-correction commit's tables; they are deliberately *not* in the claim ledger, because the ledger may only cite canonical artifacts and those bytes are superseded:

quantity	retired convention	corrected convention
`Gamma_sensitivity_matched`, 0 to 1	0.576	2.486
`Gamma_sensitivity_matched`, 1 to 2	0.387	2.120
resolved trace information	—	8.944e+08
direct trace information	—	8.845e+08
resolved-over-direct trace information gain	82%	1.12%
equalized-arm depth advantage	one grid step	tens of M

The first row is the one that mattered scientifically. Under the retired convention the sensitivity exponent was roughly seven to ten times smaller than the throughput exponent, which reads as "information decays far more slowly than flux". Corrected, it is about half — a real effect, but an ordinary one. The 82% trace-information gain becomes 1.12%: the resolved stack collects almost no additional photons, and its value is that its operational rank still rises from 153 to 201 of 224. Structure, not signal.

What did not move under the correction: the ordering of the mechanism decomposition, the negative control's superior conditioning, and full algebraic rank on C224.

The mechanism conclusion was nonetheless weakened, by a separate finding rather than by this correction. Amendment 002 (section 5.2) established that the registered mechanism statistic could not see the delay-only substitution at all. On the diagnostic that can, delay diversity is still the larger contribution but no longer the whole of it. Two distinct problems, corrected independently, both in the direction of a smaller claim.

10. Limitations

1. **Declared classes, not the continuum.** Every rank and nullity is relative to a finite-dimensional class. Section 7 is a demonstration that this qualification is load-bearing, not a formality.
2. **Three image orders.** Orders above n = 2 are not computed. No asymptotic exponent in n is fitted, and none should be read off the three points here.
3. **One emissivity and flow prescription.** The throughput exponent is a property of the frozen specific-intensity and Keplerian-flow prescription. It is not the geometric Kerr critical exponent and is not identified with it anywhere.
4. **A source domain that moves with spin.** The primary radial support follows the rays. Section 6.2 measures the confound on three anchors; it does not remove it from the primary surface.
5. **5 matched-support cells are undefined.** At low inclination the n = 0 and n = 1 retarded windows are disjoint. Those cells are reported and retained in every denominator, but they carry no exponent.

6. **No real data, no reconstruction, no learning.** The operator is analysed; nothing is inverted against an observation, and no estimator is trained.
7. **Single-instrument idealisation.** One monochromatic, spatially resolved intensity observation with white noise of fixed density per unit solid angle. No interferometric sampling, calibration, or polarisation.

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11. Reproducibility and governance

The campaign is gate-driven. Gates carry a mechanical status that is never edited to match a later adjudication, so a defect that was real stays visible:

```
gates passing:              121
active blocking failures:   0
preserved literal failures: 7
future-phase not run:       11
```

A preserved literal failure is a FAIL that has been ruled on and kept on the record rather than reinterpreted. There are 7:

preserved failure	adjudicated disposition
`EDGE1_a000_i020_raymap_generation`	`RESOLVED_BY_S0_BACKEND`
`G10q_retired_flat_sigma_convention`	`RETIRED_PIXELIZATION_DEPENDENT`
`G1_v01_reproduction_relative`	`FAIL_AS_WRITTEN`
`G7_grid_convergence_a098_i075`	`RETIRED_NONCONVERGENT_EXTREME_STATISTIC`
`G7_grid_convergence_raw_max`	`RETIRED_NONCONVERGENT_EXTREME_STATISTIC`
`G7b_weighted_operator_discrepancy`	`WITHDRAWN_INVALID_CONVERGENCE_METRIC`
`GRID_AUTHORIZATION`	`SUPERSEDED_GRID_COMPLETE`

Reproduction path: the registry digest is `2ba66f0209fe1cdec97b8cf5862494c22fb94704318f9601a7a1f9eb4b783796` ; the operator grid freeze pins the ray-map hashes, source class, probe, observer sampling, age grid, noise convention, SNR grid, arms, rank conventions, thresholds, censoring rule and permutation seeds before the first geometry was evaluated; the canonical freeze lists the 265 artifacts the manuscript may cite; and the claim ledger maps every number above to the artifact, row filter and column it came from. `scripts/verify_manuscript.py` re-derives all of them from the frozen bytes and checks that each rendered value appears in this text.

Data and code availability

All code, ray maps, tables, gate records, freezes, amendments and the invalidation ledger are in the campaign repository at tag `paper-I-campaign-final` , commit `03b0d1c9c63131aa553904df0c09849d641dee8f` . Ray tracing uses two independent tracers, pinned by version and commit, with an explicitly validated Schwarzschild backend at zero spin.